When Does Memory Help? A Cost-Aware Evaluation of Long-Term Memory in Tool-Using LLM Agents

Shweta Mishra
Independent Research
1196shweta@gmail.com

Shashank Mishra
Independent Research
24shashankm@gmail.com

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Abstract

Long-term memory for LLM agents is evaluated today by conversational recall benchmarks (LoCoMo, LongMemEval), which measure question answering over dialogue history, not whether remembered facts change what a *tool-using* agent does. We present MERIT (Memory Evaluation for Realistic Instrumented Tasks), a benchmark and harness that measures the *marginal utility* of memory for task-executing agents under explicit cost accounting. MERIT provides episodic tool-use tasks in three domains whose dependence on earlier-episode facts is verified by an automated leak check; a difficulty ladder ending in *updated-fact* recall; controlled memory corruption; and full token and dollar metering of every memory operation. Across 23,440 scored episodes ($42.57), a two-generation pilot on `gpt-4.1-mini` and a preregistered 3-model × 3-seed grid (GPT-4.1, Claude Haiku 4.5; memory side held fixed), memory lifts dependent-task success from a leak-verified floor of 0.00 to 0.55–1.00. On updated facts, embedding retrieval collapses *unpredictably* (0.30–0.95 across models; max seed gap 0.45), and agents act on a correctly retrieved value only 55% of the time, while update-on-write stores (a structured fact store and, notably, LLM summarization) remain at 0.70–1.00; the hybrid is worse than the fact store alone. A latest-generation spot-check (Claude Sonnet 5, gated on a clean full-replay control) reproduces the pattern. Swapping a memory's implementation moves task success by up to 60 points, and full replay is never economical: the best condition per domain delivers 2.7–3.9× its marginal utility per dollar. We release the benchmark, harness, and all traces.

1 Introduction

LLM-based agents that plan, call tools, and act over multiple sessions are moving rapidly into production: customer support, IT operations, coding, and personal assistance. A central architectural question for every such deployment is whether and how to give the agent *long-term memory*: information persisted across sessions and selectively recalled. A vibrant ecosystem of memory systems has emerged (Mem0, Letta/MemGPT-style OS memories, Zep, LangGraph checkpointing, bespoke RAG stores), and vendors compete on recall benchmarks.

Yet the evaluation of agent memory has three blind spots:

*All numbers in §5 are measured results: a two-generation pilot (9,940 scored episodes, `gpt-4.1-mini`), run first with starter memory implementations and then real ones (embedding retrieval, LLM summarization, LLM extraction) on the identical grid, followed by the preregistered full grid (§5.7): 3 agent models × 3 domains × 3 difficulty tiers with 3 seeds on the primary model, 13,500 further episodes. Pilot and full grid: 23,440 scored episodes, $42.57 in API cost; latest-generation and frontier spot-checks (Claude Sonnet 5, Opus 4.8) are reported separately as diagnostics in §7. Code, traces, and preregistration: `https://github.com/smshweta/merit-bench`.

Blind spot 1: Conversational QA is not task execution. The dominant benchmarks (LoCoMo (Maharana et al., 2024), LongMemEval (Wu et al., 2025), and BEAM (Tavakoli et al., 2025)) test whether a system can answer questions about long, multi-session *conversations*. Production agents, by contrast, must use remembered facts to *choose the correct tool call*: the right customer ID, the previously agreed refund amount, the configuration fix identified during last week's incident. Whether conversational recall scores transfer to task-execution utility is an open empirical question.

Blind spot 2: Memory is assumed to help; harm is unmeasured. Recent work notes that agents sometimes ignore retrieved memory records or use them inconsistently, and HaluMem (HaluMem, 2025) examines memory hallucination and consistency, but in conversational settings. In task execution, a stale memory (a customer's *old* address, a rolled-back configuration) is not just unhelpful; it produces a *confidently wrong action*. No existing benchmark quantifies this failure mode under controlled corruption, nor the more basic failure our pilot surfaces: agents that demonstrably hold the correct fact in context and still do not act on it (Ignore Rate 0.45–0.53 across implementations).

Blind spot 3: Cost is reported, but marginal utility is not. Memory systems report token counts alongside accuracy, but the decision practitioners face is economic: does adding memory raise task success enough to justify its per-task cost? We are not aware of any evaluation that reports *cost-adjusted marginal utility* (the change in success probability per marginal dollar) across memory architectures.

We address all three with MERIT. Our contributions:

1. **A task suite with controllable memory dependency and difficulty.** Episodic tool-use tasks in three domains (customer support, IT operations, personal assistant) where a tunable fraction of episodes depend on facts established in earlier episodes, and a three-tier difficulty ladder: *easy* (one fact), *medium* (multiple facts must be composed), *hard* (the fact is updated mid-arc and the latest value is required). An automated **leak check** verifies at generation time that no gold fact is present in the probe episode's inputs or re-derivable from tools.

2. **Controlled memory corruption.** Stale, contradictory, and distractor records injected at known rates ρ, with ground-truth flags, to measure Stale-Memory Harm.

3. **Three metrics absent from prior evaluations**: Memory Utilization Rate (MUR), Ignore Rate, and Cost-Adjusted Marginal Utility (CAMU), computed automatically by the harness (§3.5).

4. **A two-generation pilot study** of 6 memory conditions $\times$ 3 domains $\times$ 3 difficulty tiers (9,940 scored episodes; the full grid run twice, with starter implementations and with real ones) with preregistered hypotheses and statistics (paired bootstrap clustered by arc; Holm–Bonferroni), demonstrating that the benchmark separates memory architectures that are indistinguishable under single-fact recall, and separates *architecture* from *implementation quality*; plus a 3-agent-model $\times$ 3-seed replication (13,500 further episodes, §5.7) showing the updated-fact collapse of embedding retrieval is model- and seed-unstable while update-on-write memories are uniformly robust.

5. **Open-source release** of the benchmark, harness, and all traces, with a deterministic $0 mock-model mode that validates the full pipeline.

2 Related Work

2.1 Memory systems for LLM agents

MemGPT (Packer et al., 2023) frames memory as an OS-style hierarchy with the LLM paging information in and out of context. Mem0 (Chhikara et al., 2025) extracts and consolidates structured facts at write time; Zep (Rasmussen et al., 2025) maintains a temporal knowledge graph; A-Mem (Xu et al., 2025) organizes agentic notes dynamically. Rolling summarization is the default pattern in agent frameworks such as LangChain/LangGraph. These systems papers evaluate primarily on conversational QA (LoCoMo, LongMemEval); none evaluate task-execution utility under cost accounting. MERIT's conditions C1–C5 are deliberately *reimplementations of these architecture families* under one interface, so that architectural mechanisms (retrieval, summarization, update-on-write, hybrid) can be compared in isolation from product engineering. We deliberately evaluate reimplementations rather than the shipped products: a product's score conflates its architectural mechanism with proprietary retrieval tuning, prompting, and engineering, preventing causal attribution to the mechanism; benchmarking shipped systems on MERIT is a natural follow-up enabled by the released harness.

2.2 Benchmarks for agent memory

LoCoMo (Maharana et al., 2024) evaluates QA over very long multi-session dialogues. LongMemEval (Wu et al., 2025) covers information extraction, multi-session reasoning, temporal reasoning, knowledge updates, and abstention over ∼115K-token histories. BEAM (Tavakoli et al., 2025) scales conversational probing to 10M tokens. HaluMem (HaluMem, 2025) evaluates hallucination at the level of memory operations (extraction, updating, QA). RealMem (Bian et al., 2026) moves toward project-oriented interaction. All are *answer-producing* evaluations; MERIT is *action-producing*: success is a predicate over the final state of a mutable world, and the knowledge-update dimension that LongMemEval and HaluMem probe conversationally becomes, in MERIT's hard tier, a behavioral test of whether the agent *acts* on the latest value. Long-context benchmarks (RULER, Hsieh et al., 2024; BABILong, Kuratov et al., 2024) test single-pass attention over a given context, not the write-and-retrieve loop of a persistent memory.

2.3 Tool-use and agent benchmarks

τ-bench (Yao et al., 2024) evaluates tool-calling agents conversing with simulated users over domain APIs with database-state success checks (the closest environment design to MERIT's), but each task is a single episode; memory across episodes is not exercised. AgentBench (Liu et al., 2023), SWE-bench (Jimenez et al., 2024), and WebArena (Zhou et al., 2023) similarly evaluate within-episode competence. MERIT adds the cross-episode dependency structure (arcs, plants, probes, leak check) on top of a τ-bench-style environment.

2.4 Reliability of agents

Production reports consistently rank reliability as the top barrier to agent deployment; confidently-wrong actions are costlier than abstentions. Stale-memory harm is a concrete, measurable instance: MERIT injects staleness at known rates and measures both the success drop and whether the agent re-verifies before acting.

3 The MERIT Benchmark

3.1 Design goals

G1 (Ecological validity): tasks require tool calls with programmatically verifiable end states. G2 (Controllable dependency and difficulty): the dependent-task ratio and the difficulty tier are benchmark parameters. G3 (Adversarial realism): corruption reflects real staleness processes. G4 (Cost transparency): every token in and out of the memory system is metered. G5 (Reproducibility): deterministic seeded worlds, scripted or cached simulated users, pinned model versions, released traces, and a $0 deterministic mock model that exercises the entire pipeline.

3.2 Environment

An **episode** is one agent session with a simulated user and a tool API over a mutable world state (SQLite). Episodes are grouped into **arcs** of 4–6 episodes sharing entities. Arcs contain **plant** episodes (a fact is established conversationally; the user explicitly forbids acting on it yet), optional **update** episodes (the fact is revised), **probe** episodes (success requires the fact; it is absent from the episode's inputs), and **independent** episodes (solvable within-episode).

Two safeguards proved essential in practice. The **leak check** asserts at generation time that every gold fact value is absent from the probe episode's inputs and from the initial world state, with gold values kept unique arc-wide. **Delta scoring** marks an episode successful only if its checker predicate flips from false to true *during* that episode: in early gate runs, eager agents processed refunds during plant episodes, and probes then "succeeded" off inherited world state; delta scoring eliminates this world-state leak channel entirely (across 9,940 episodes, the 22 probes that arrived with their checker already satisfied score as failures, never as inherited successes).

3.3 Domains

- **D1 Customer support (retail):** tools = `get_order`, `refund`, `update_address`, `get_policy`, `send_message`. Facts: agreed partial-refund amounts, updated addresses.
- **D2 IT operations:** tools = `search_logs`, `get_deploy_history`, `get_config`, `set_config`, `deploy`, `update_ticket`. Facts: diagnosed config fixes, safe rollback versions.
- **D3 Personal assistant:** tools = `get_calendar`, `create_event`, `send_email`, `get_preference`, `set_preference`. Facts: standing room/time preferences, dinner commitments (place, day, time).

3.4 Difficulty ladder and memory conditions

Difficulty: *easy* = one gold fact per probe; *medium* = multi-fact probes (every gold value must appear in the executed tool calls); *hard* = updated-fact probes: the value is planted, revised in a later episode, and the probe requires the **latest** value. The hard tier is a targeted stressor for the architectural distinction between stores that overwrite (update-on-write) and stores that accumulate (replay, retrieval).

Memory conditions, one interface (`write(episode)`, `read(context)`): C0 none; C1 full replay of prior transcripts; C2 retrieval (starter: keyword-overlap; real: embedding retrieval, `text-embedding-3-small`); C3 rolling summary (starter: extractive truncation; real: LLM summarization); C4 structured fact store with update-on-write (starter: pattern-based extraction; real: LLM extraction); C5 hybrid (C4 + C2). Each condition ran in both an inexpensive **starter** implementation and a **real** one (memory-side LLM/embedding calls, fully metered); the grid was run once per generation (§5.3 compares them). All conditions share the

same agent scaffold (ReAct-style tool loop; Yao et al., 2023), prompts (except the memory block), tools, and decoding (temperature 0).

3.5 Metrics

- **TSR**: task success rate by programmatic end-state check, split by dependent/independent.
- **MUR**: among dependent episodes where all gold values were present in the retrieved memory block, the fraction where every gold value appears in the executed tool-call arguments (value tracing). A frozen, stratified 100-episode audit sample is committed to the repository. Two human annotators labeled all 100 items blind to the tracer's output: the tracer agrees with the first on 93% of items (Cohen's $\kappa = 0.63$, substantial agreement) and with the second on 93% of the 99 items not marked unclear ($\kappa = 0.60$), and the annotators agree on all 99 items both labeled ($\kappa = 1.00$). All seven disagreements with each annotator are in the same direction, with the tracer scoring *not utilized* where the humans judged the value was used (typically reformatted values the containment check misses), so the tracer under-counts utilization and the Ignore Rates we report are conservative upper bounds.
- **Ignore Rate** = 1 − MUR on episodes with correct memory present.
- **SMH**: TSR(clean) − TSR(corrupted) at $\rho \in \{0.1, 0.3\}$ for stale / contradiction / distractor corruption.
- **Cost & CAMU**: metered tokens and dollars per episode; CAMU = ΔTSR vs. C0 per Δcost vs. C0; break-even task value = Δcost/ΔTSR.

3.6 Statistical methodology

Paired comparisons on identical task instances; 95% CIs and two-sided p by paired bootstrap (10,000 resamples) clustered at the arc level; Holm–Bonferroni within each preregistered hypothesis family (H1–H4, committed to the public repository before experiments; commit history is the preregistration record).

4 Experimental Setup

Model: `gpt-4.1-mini` (pinned; temperature 0) on both the agent side and, in the real-implementation generation, the memory side (embeddings: `text-embedding-3-small`). Scale per generation: 10 arcs × 5 episodes per (domain × difficulty × condition); dependent-task ratio 0.5; corruption sweep (3 modes × $\rho \in \{0.1, 0.3\}$) with LLM-paraphrased users on D1. Totals: **9,940 scored episodes, $9.12 in API cost**; starter generation 5,440 episodes / $4.61, real generation 4,500 episodes / $4.52 (~$0.001/episode including memory-side calls, which are metered into all cost figures). Harness: Python, LiteLLM for provider-agnostic calls and metering; SQLite worlds; scripted simulated users (LLM paraphrase mode with a paraphrase-leak check on D1). The mock-model mode replays the entire grid deterministically at $0 and is exercised by 62 unit tests, including generation determinism, leak checks, and end-to-end pipeline invariants. Unless marked otherwise, §5.1–5.6 report the real-implementation generation on `gpt-4.1-mini`.

Full grid (§5.7). The preregistered multi-model grid runs the real implementations on three agent models: `gpt-4.1-mini` (3 seeds), GPT-4.1 (frontier, 1 seed), and Claude Haiku 4.5 (cross-vendor, pinned `claude-haiku-4-5-20251001`, 1 seed), over the identical 3-domain × 3-tier grid, 10 arcs × 5 episodes per cell. The memory side is pinned to `gpt-4.1-mini` (embeddings: `text-embedding-3-small`) for *every* agent model, so the agent model is the only varying factor. Totals: **13,500 scored episodes, $33.45** ($20.46 OpenAI + $12.99 Anthropic), bringing the study to 23,440 episodes and $42.57 overall.

Table 1: Dependent-task TSR by difficulty tier and memory condition, real implementations (D1 / D2 / D3). See also Figure 1.

Tier	C1 replay	C2 retrieval	C3 summary	C4 facts	C5 hybrid
easy	1.00 / 1.00 / 0.90	0.90 / 1.00 / 0.90	0.95 / 1.00 / 0.70	1.00 / 0.55 / 0.80	0.95 / 0.80 / 1.00
medium	0.60 / 1.00 / 0.95	0.30 / 0.95 / 1.00	0.30 / 1.00 / 0.90	0.80 / 0.40 / 0.75	0.80 / 1.00 / 0.95
hard	1.00 / 0.95 / 1.00	0.70 / 0.35 / 0.45	1.00 / 0.70 / 1.00	0.95 / 0.75 / 1.00	0.80 / 0.60 / 0.50

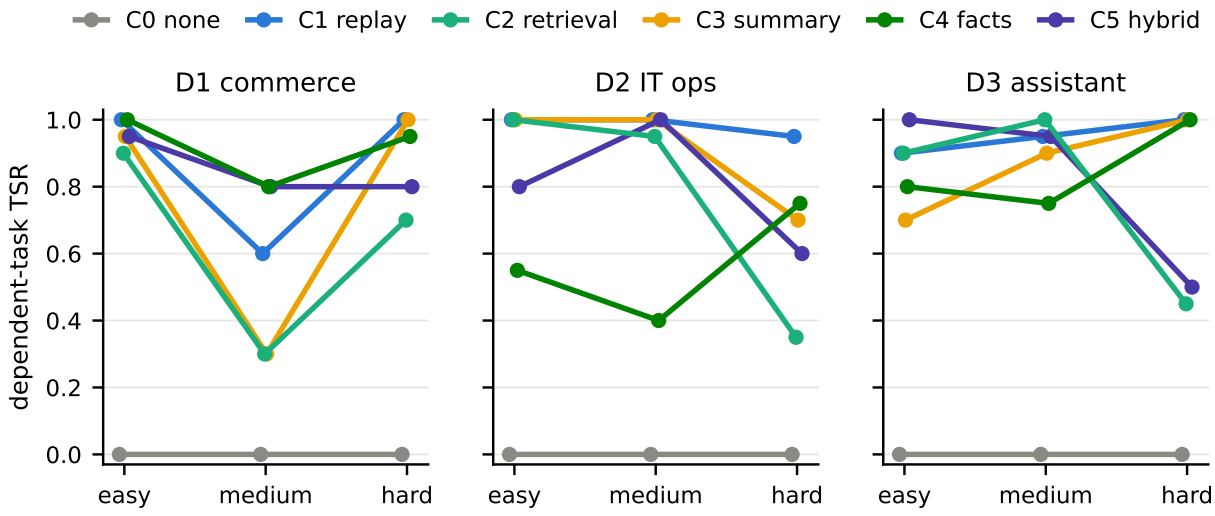


Figure 1: Dependent-task TSR across the difficulty ladder, per memory condition and domain (real implementations). The hard (updated-fact) tier separates memories that overwrite state, i.e., the fact store (C4) and, notably, LLM summarization (C3), which rewrites its summary each episode, and chronological replay (C1) from retrieval-based memories (C2, C5).

5 Results

5.1 Memory helps on dependent tasks; the floor is real

C0 scored **0.000** on dependent tasks in all nine (domain × difficulty) cells, in both generations: the leak check holds; there is no route to the gold facts except memory. On independent tasks C0 scores 0.83–1.00, confirming task solvability. Every memory condition beats C0 on dependent tasks in every domain at easy difficulty (ΔTSR +0.55 to +1.00; all Holm-adjusted $p \leq 0.001$, paired bootstrap clustered by arc).

5.2 The difficulty ladder dissociates architectures

Table 1 and Figure 1 show three dissociations: (1) **Updated facts break retrieval memory, including embedding retrieval**: C2 falls to 0.35–0.70 on hard while C1 stays at 0.95–1.00 and C4 at 0.75–1.00: retrieval surfaces stale and fresh values side by side with no recency signal, while update-on-write overwrites and replay resolves recency from chronology. (2) **Hybrid is worse than its better half** on hard in all three domains (0.50–0.80 vs. C4's 0.75–1.00): the retrieval half re-imports the staleness the fact store had eliminated. (3) **Full replay is update-immune but composition-limited**: C1 drops to 0.60 on D1-medium despite containing every fact: possessing information and composing it are different capabilities. A fourth pattern emerged only with real implementations: **LLM summarization behaves like update-on-write** (1.00

/ 0.70 / 1.00 on hard), because regenerating the summary each episode naturally keeps the latest value, an architectural kinship invisible in the starter generation (§5.3).

5.3 Starter vs real implementations: the swap is diagnostic

Upgrading C2–C5 in place, on the identical grid, moves conditions in both directions (Figure 2):

- **C3 summary, truncation → LLM:** 0.00 / 0.15 / 0.00 → **1.00 / 0.70 / 1.00** on hard. The starter result was a floor set by the implementation, not the architecture: truncation drops facts (starter MUR = 1.0 on the episodes it retained), while an LLM summarizer that rewrites state each episode is update-robust by construction.
- **C4 facts, patterns → LLM extraction:** D2 falls from 1.00 to **0.40** on medium (0.60 → 0.55 easy, 0.90 → 0.75 hard) while D1/D3 roughly hold. The hand-tuned patterns the starter needed per domain (§6) were not dead weight; the generic LLM extractor misses IT-ops facts they caught. Extraction quality is a real tax on structured memory, payable in either engineering effort or metered tokens and now visible in the benchmark.
- **C2 retrieval, keyword → embedding:** modest gains everywhere (+0.10 to +0.25 on hard), but the hard-tier collapse persists (the failure is the architecture's missing recency arbitration, not retrieval quality).

The two generations bound an implementation-sensitivity band per architecture; the band is wide (up to 1.00 TSR for C3 on updated facts), which is itself an argument for action-level evaluation over memory-system benchmarking.

5.4 Agents ignore memories they hold

On the hard tier, pooling domains, the correct (latest) value was present in C2's retrieved block in 55 probe episodes; the agent acted on it in 30 (**Ignore Rate 0.45**; 0.53 in the starter generation; upgrading retrieval quality barely moves it). Even C1, whose memory block is a clean full transcript, ignores up to 0.50 of held facts in multi-fact (medium-tier) episodes (Figure 3). This is Blind spot 2 made measurable: memory-system accuracy overstates end-task benefit unless utilization is measured.

5.5 Stale-memory harm

On D1 with stale corruption (Figure 4), the largest and only Holm-significant harm is again on the hybrid C5 (SMH +0.25 at $\rho = 0.3$, Holm-adjusted $p = 0.030$), replicating the starter-generation result with real implementations. One shift is instructive: C4 with LLM extraction shows stale harm of +0.20 (raw $p = 0.013$, not surviving Holm) where regex-C4's harm was ≤ 0.05: the LLM extractor ingests corrupted records that the rigid patterns rejected, another face of the extraction-quality tax (§5.3). Contradiction and distractor corruption produce small, non-significant effects at pilot scale. (Corruption sweeps beyond D1, with verification-rate analysis, are left to future work.)

5.6 Cost and marginal utility

With memory-side calls metered, per-episode cost on D1-easy (Figure 5): C0 \$0.00046, C4 \$0.00066, C2 \$0.00073, C3 \$0.00095, C5 \$0.00111, C1 \$0.00126 (2,914 tokens/episode; 2.7× C0's cost). CAMU and accuracy rankings differ in every domain (H4 supported): the best CAMU is C4 in D1 (4,839 points of dependent-task success per marginal dollar) and D3 (3,245), but **C2 in D2** (6,629), where LLM extraction's

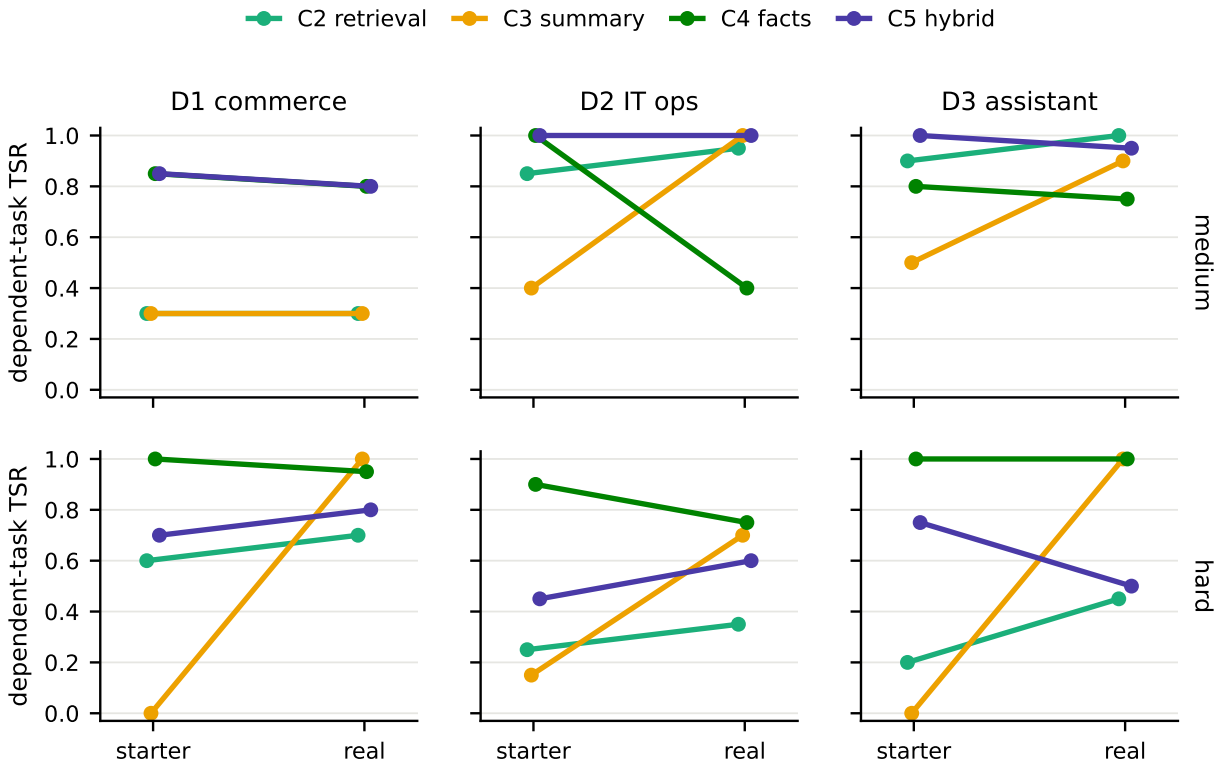


Figure 2: Dependent-task TSR, starter vs real implementation of each memory condition (C2–C5), on the medium (top) and hard (bottom) tiers. LLM summarization rescues C3 on updated facts; LLM extraction costs C4 up to 60 points in D2; embedding retrieval does not fix C2's hard-tier collapse.

accuracy regression (§5.3) erased C4's edge. Full replay is never the economical choice (1,222–1,741 pts/$; 2.7–3.9× worse than the per-domain best). Two starter-generation artifacts vanish under honest metering: C4's headline ~17,300 pts/$ (regex extraction was free) becomes 4,839, and its *negative* marginal cost in D2 becomes +$0.0003: the extraction tax, now on the books. Break-even task values remain fractions of a cent for all conditions: memory pays for itself at trivially low task values *when it works*; the practitioner-relevant differences are in robustness (§5.2–5.5), not raw affordability, at these model prices. The absolute CAMU figures are tied to mid-2026 API pricing and will drift as prices change; the qualitative result, that robustness rather than per-episode cost is the deciding factor and that full replay never leads on CAMU, is price-independent, since it rests on the ratio of success gains to cost *differences* between conditions.

5.7 The full grid: seed robustness and cross-model generality

The full grid (§4) asks two questions the pilot could not: do the pilot's dissociations survive resampling (3 seeds on `gpt-4.1-mini`), and do they survive a change of agent model (GPT-4.1; Claude Haiku 4.5, a different vendor and tier) with the memory side held fixed?

Seed variance concentrates in embedding retrieval. Across the nine 3-seed cells, the maximum pairwise TSR gap between seeds averages 0.16 for C2, the largest of any condition, with a worst cell of 0.45 (C2, D1-hard: per-seed TSR 0.45 / 0.90 / 0.75). C1 averages 0.07; the overwrite-style memories C3 and C4 average 0.11–0.12. A single-seed evaluation of embedding retrieval on the hard tier could honestly report anywhere

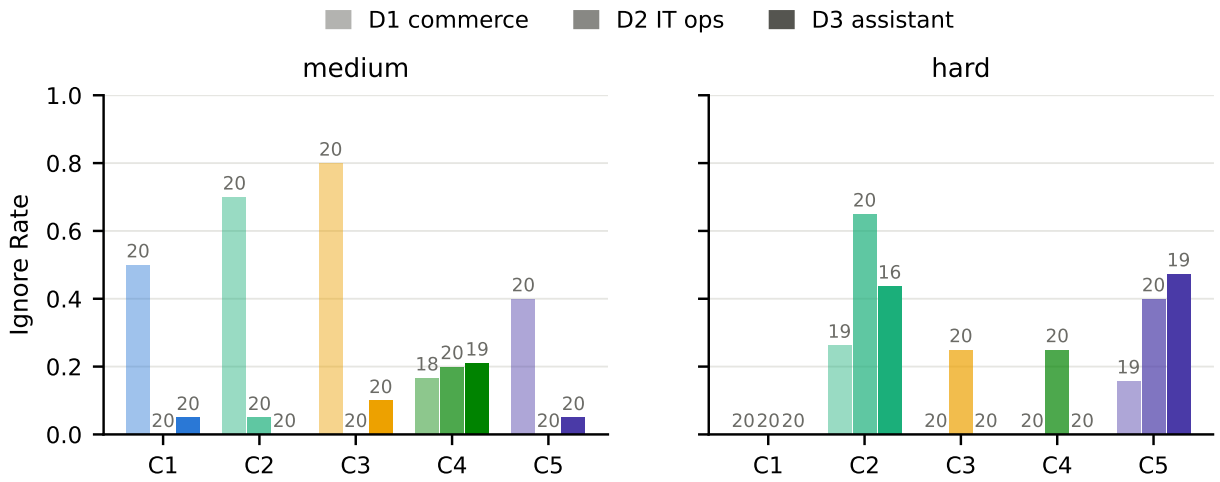


Figure 3: Ignore Rate (the fraction of dependent episodes where every gold value was present in the memory block but the agent did not act on it) by condition and domain on the medium and hard tiers (real implementations). Numbers above bars are episode counts with memory present (bars at zero are shown by their count only).

from 0.45 to 0.90 for the same system; seed-mean numbers with spread are the only defensible summary, and all numbers below are seed means.

The hard-tier collapse replicates across models: a reliability failure, not a fixed deficit. Table 2 and Figure 6 report the hard (updated-fact) tier for all three models. Embedding retrieval (C2) is the only architecture that collapses, but *where* it collapses is model-idiosyncratic: Haiku 4.5 holds the single-fact domains (0.90 / 0.95 on D1 / D2) where GPT-4.1 and `gpt-4.1-mini` drop to 0.38–0.70, yet Haiku falls hardest on multi-fact D3 (0.30). No model escapes: every model has at least one domain at $\leq$0.60, and the spread across model $\times$ domain is 0.30–0.95. LLM summarization (C3), by contrast, spans 0.80–1.00 on the same cells (robust for every model, domain, and seed), and the structured fact store (C4) spans 0.70–1.00.

The dissociation is significant for every model. Pooling domains on the hard tier, the paired arc-clustered bootstrap gives C3$-$C2 = +0.44 [+0.36, +0.53] for `gpt-4.1-mini`, +0.38 [+0.23, +0.53] for GPT-4.1, and +0.28 [+0.15, +0.43] for Haiku 4.5 (all $p \leq 0.0002$; C4$-$C2 positive for all, $p \leq 0.009$), and +0.25 [+0.10, +0.40] ($p = 0.002$) for the Sonnet 5 spot-check cell. The per-domain contrasts localize Haiku's entire deficit in D3 (+0.70 [+0.45, +0.95]; D1/D2 not significant): the significance structure itself traces the model-idiosyncratic locus of collapse.

The refined claim is stronger than the pilot's: on updated facts, embedding retrieval is not merely worse: it is **unpredictable**, both across seeds (max gap 0.45) and across agent models (0.30–0.95), because success depends on whether that particular agent resolves the stale-versus-fresh conflict in retrieved context, an ability that varies non-monotonically with model tier. Update-on-write architectures remove the conflict at write time and are correspondingly stable everywhere. A practitioner choosing a memory system from a single-model, single-seed benchmark number would systematically over- or under-estimate embedding retrieval; MERIT's grid makes the variance itself measurable.

A latest-generation check. As a final probe we ran the D1-hard cell on Claude Sonnet 5, a 2026-generation agentic model, gated on the same C1 control that disqualified Opus 4.8 (§7): here the control is clean (C1

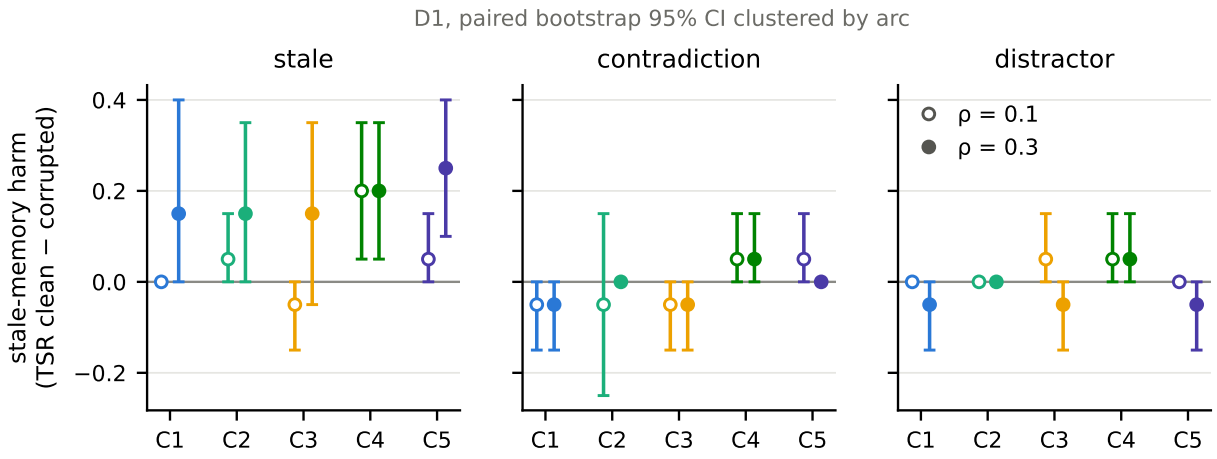


Figure 4: Stale-memory harm (TSR clean − corrupted, dependent tasks, D1, real implementations) by condition, corruption mode, and corruption rate ρ; error bars are paired bootstrap 95% CIs clustered by arc.

hard tier, TSRdep		Haiku 4.5	GPT-4.1	gpt-4.1-mini (±sd)
C2 embedding retrieval	D1	0.90	0.70	0.70 ± 0.23
	D2	0.95	0.45	0.38 ± 0.08
	D3	0.30	0.60	0.37 ± 0.03
C3 LLM summarization	D1	1.00	1.00	0.98 ± 0.03
	D2	1.00	0.90	0.80 ± 0.10
	D3	1.00	1.00	1.00 ± 0.00
C4 structured fact store	D1	1.00	1.00	0.93 ± 0.08
	D2	1.00	1.00	0.90 ± 0.05
	D3	0.70	0.90	0.90 ± 0.09

Table 2: Hard (updated-fact) tier across agent models, memory side pinned to `gpt-4.1-mini`. C0 floor is 0.00–0.05 and C1 full replay is 1.00 for every model (omitted). Full per-cell tables for all tiers are in the released traces.

= 1.00, C0 floor 0.00). The pattern replicates: embedding retrieval (C2) drops to 0.75 while LLM summarization (C3) and the fact store (C4) hold 1.00 (hybrid C5 0.90). The updated-fact weakness of retrieval-only memory is not an artifact of older model generations.

Each real implementation remains one representative of its family (one embedding model, one summarization prompt, one extraction prompt), and §5.3 shows how much such choices matter.

6 Discussion

Practitioner guidance (provisional). If tasks depend on facts that get *revised* (addresses, configs, schedules; that is, most operational facts), prefer a memory that *overwrites state* (a structured fact store, or LLM summarization, which turns out to be update-robust because it rewrites its summary every episode) over retrieval, which accumulates. Do not assume a hybrid inherits the better component's behavior: measure it. Full replay is a strong accuracy baseline that fails on cost (2.7× tokens) and on multi-fact composition. And treat the write path as a first-class risk: swapping extraction implementations moved C4 by 60 points in one domain (§5.3); truncation is not summarization.

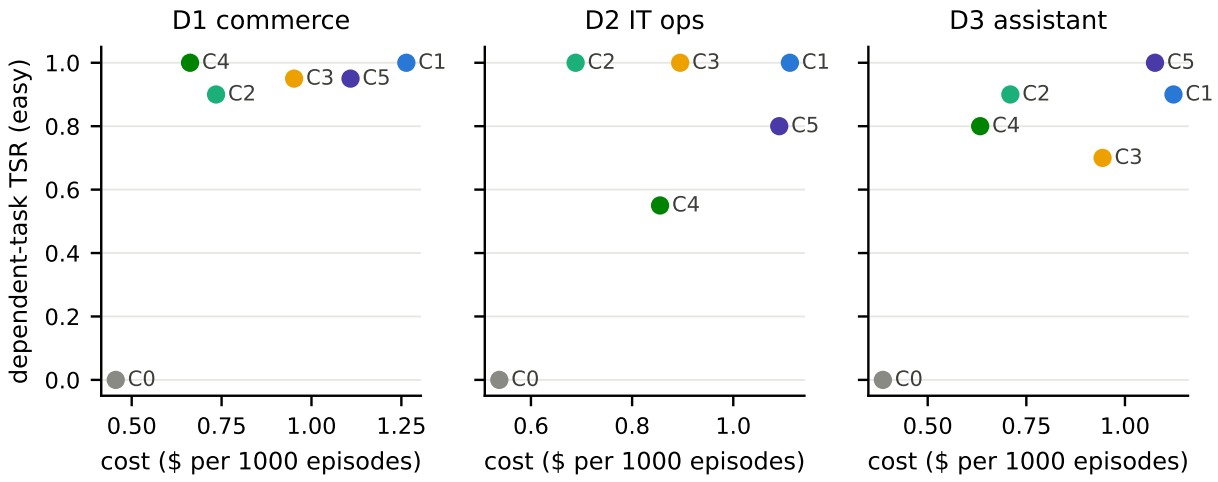


Figure 5: Metered cost per episode (including memory-side calls) vs. dependent-task TSR at the easy tier, real implementations. C4 stays on the Pareto frontier in D1/D3 but cedes it in D2; C1 full replay pays a 2–3× cost premium for equal or lower TSR everywhere.

Why do agents ignore correct memories? Traces show two patterns: (i) under retrieval, stale and fresh values co-occur and the agent averages, asks, or picks the stale one (no provenance/timestamps to arbitrate); (ii) under multi-fact composition, agents act on the subset of facts nearest the task phrasing. Both suggest memory *presentation* (provenance, recency marking, contradiction surfacing) matters as much as memory *storage*.

Extraction brittleness as a hidden cost, now measured. In the starter generation, C4's pattern extractor was blind outside its home domain until hand-tuned patterns were added; the real generation shows the converse: generic LLM extraction pays for its generality with a 60-point accuracy regression in D2 and greater willingness to ingest stale records (§5.5). In production, extraction quality is the tax that structured memory pays, payable in domain engineering or in metered tokens, but never zero. MERIT puts it on the books (CAMU includes memory-side calls).

A methodological note. Two of our safeguards were added because early runs failed without them: eager agents leaked gold facts into world state (fixed by delta scoring), and gold-value collisions leaked across task pairs (fixed by arc-wide uniqueness). Task-execution memory benchmarks have leak channels that conversational QA benchmarks cannot express; we recommend leak checks over *world state*, not just prompts, as standard practice.

7 Limitations and Threats to Validity

Construct: programmatic checkers may not capture all real-world success notions; the MUR tracer is string containment, validated against a two human annotators ($\kappa = 0.63$ and 0.60, conservative direction; the annotators agree on all 99 items both labeled), and the frozen audit sample is committed. **Internal:** prompt differences across conditions are confined to the memory block; delta scoring removes world-state carryover; the mock-model grid guards the pipeline, but mock results are never reported as findings. **External:** the grid spans three agent models plus two gated spot-checks, and the spot-checks bound the model-strength question

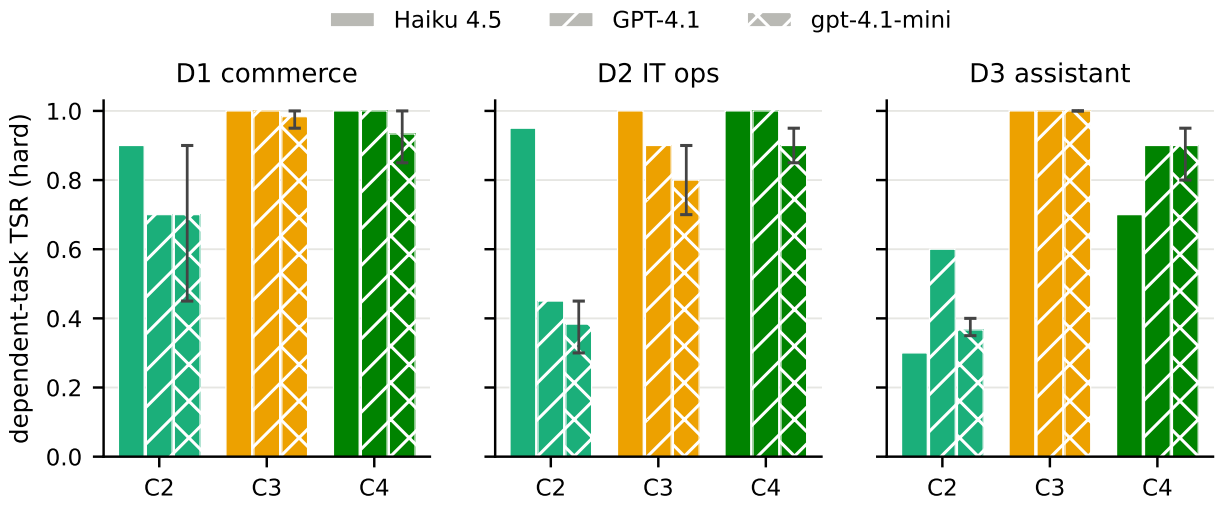


Figure 6: Hard-tier TSR across agent models (bar hatch = model, color = condition; whiskers = min–max across the 3 `gpt-4.1-mini` seeds). Embedding retrieval (C2) collapses somewhere for every model, but where is model-idiosyncratic; LLM summarization (C3) is robust everywhere.

in both directions: frontier capability does not substitute for memory (C0 = 0.00 even for Claude Opus 4.8 and Sonnet 5), and the C2 collapse persists on the newest generation. Yet all findings remain within three synthetic domains with scripted users (LLM-paraphrase mode mitigates phrasing overfit on D1), 10 arcs per cell, and each real memory implementation is a single representative of its family; §5.3 quantifies how consequential implementation choices are. **Reproducibility:** deterministic seeded generation, pinned model IDs, released traces, $0 mock mode. API model deprecation remains a limitation: the released traces preserve the reported runs, and the harness accepts any OpenAI-compatible endpoint (including locally served open-weight models), so the grid is rerunnable even after the reported API models retire.

Frontier-model spot-check: task success conflates memory use with policy prudence. After the full grid we ran one diagnostic cell (D1-hard, 6 conditions × 50 episodes) on Claude Opus 4.8, the strongest model available to us. The no-memory floor held (C0 = 0.00): frontier capability cannot substitute for memory on leak-verified dependent tasks. But the full-replay control collapsed to C1 = 0.75 (vs 1.00 for all three grid models), and the transcripts show why: on updated-fact probes the model *quotes the remembered amounts and then declines to act on them*, observing that the update trail "traces back only to my own confirmation messages" repeating user-asserted values, and requesting supervisor confirmation for what it reads as a chat-escalated refund, arguably correct behavior for a production agent, and plausibly a product of safety training. Two implications. First, raw TSR comparisons across models conflate memory use with policy prudence; the C1 control detects exactly this confound, and cross-model numbers should be read relative to each model's own C1. Second, MERIT's updated-fact arcs (a user renegotiating an amount across sessions) are structurally similar to social-engineering escalations, and safety-tuned models may increasingly treat *memory provenance* as part of the decision. Benchmarks that score only task completion will under-credit such models; we report this cell as a diagnostic, not a leaderboard entry. By contrast, Claude Sonnet 5, the same vendor's current agentic-tuned model, passes the control cleanly (C1 = 1.00) and reproduces the pattern of §5.7 (C2 0.75 vs C3/C4 1.00 on D1-hard): the prudence confound is a property of specific safety postures, not of model generation, and the C1 control suffices to detect it per model.

We flag this as more than a measurement nuisance. It suggests a distinct axis that action-level memory

evaluation must contend with as models are increasingly safety-tuned: *memory-provenance skepticism*, in which an agent that correctly recalls a value nonetheless declines to act because it cannot establish that the value was legitimately authorized rather than user-asserted. On MERIT's updated-fact arcs this manifests as a lower ceiling on TSR that is not a memory failure at all (the fact is retrieved and quoted) and that a naive leaderboard would misread as a weaker agent. Whether the behavior is desirable (it plausibly prevents social-engineering in production) or over-cautious (it blocks legitimate task completion) is context-dependent and beyond our scope; our claim is narrower: the C1 full-replay control isolates it from genuine memory failure, so cross-model TSR should always be read relative to each model's own C1. This observation rests on a single diagnostic cell (one domain, 50 episodes) and two frontier models; characterizing provenance skepticism across domains, prompts, and model families is left to future work, but MERIT's control structure already makes it visible and separable, which we believe is a prerequisite for studying it at all.

8 Conclusion

MERIT reframes agent-memory evaluation from "can the system recall?" to "does recall change what the agent does, at what cost, and how does it fail?". The answer is not monotone: architectures indistinguishable on single-fact recall separate by up to 0.70 TSR when facts must be superseded (C3 1.00 vs. C2 0.30, Haiku 4.5, D3-hard), agents ignore nearly half of correctly retrieved facts, swapping one memory implementation for another moves task success by as much as 1.00 TSR in either direction, and the most accurate memory is 2.7–3.9× less economical than the most efficient one. The benchmark, harness, traces, and preregistration are public; we invite memory-system authors to evaluate against MERIT.

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